

Preclass Learning: Code and Data Quality Control (ICA 19)

1. Code Quality and Best Practices

- **Readability:** Adhering to a consistent **Style Guide** (e.g., for variable names, spacing, and function length) to ensure other developers can understand the code easily.
- **Modularity:** Using functions to prevent code duplication (DRY principle) and organize logic.
- **Documentation:** Clear comments explaining *why* the code is doing something complex, and comprehensive documentation for every function (inputs, outputs, and side effects).
- **Testing (Unit Tests):** Writing small, automated tests for individual functions to ensure they reliably produce the correct output for known inputs.

2. Data Quality Assurance

- **Validation:** Implementing checks on the data after import and cleaning:
 - **Range Checks:** Ensuring values fall within a logical domain (e.g., percentages between 0 and 100).
 - **Type Checks:** Verifying that columns have the expected data type.
 - **Uniqueness Checks:** Confirming that primary keys contain no duplicate values.
- **Reproducibility:** Ensuring that the entire analysis can be rerun from the raw data to the final output, producing the exact same results every time. This requires strict version control of both code and data processing steps.